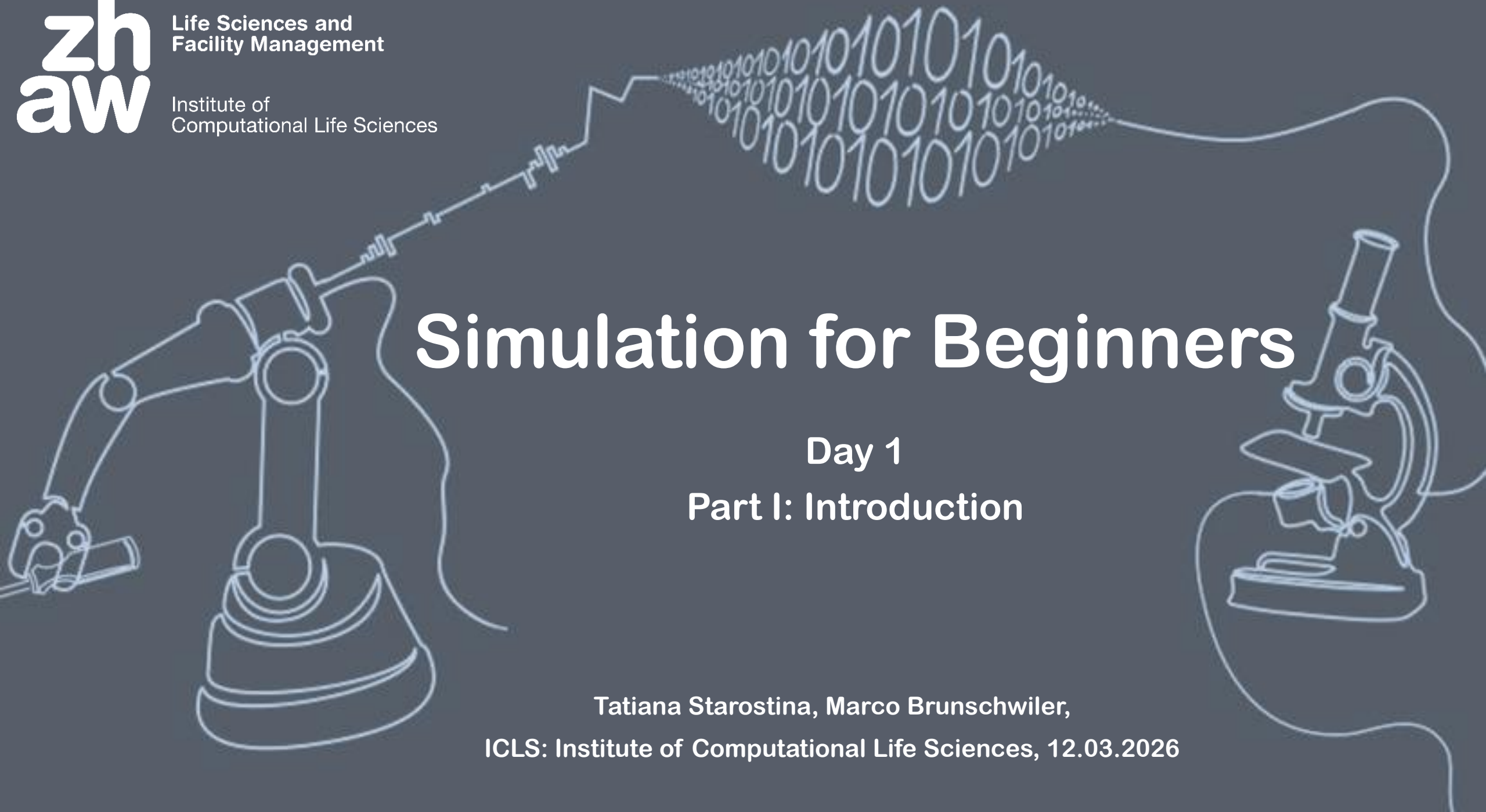
Life Sciences and
Facility Management

Institute of
Computational Life Sciences



Simulation for Beginners

Day 1

Part I: Introduction

Tatiana Starostina, Marco Brunschwiler,

ICLS: Institute of Computational Life Sciences, 12.03.2026

Agenda

Part I: Introduction and theory

- Introduction of all participants
- Organization of module
- What is simulation?
- Simulation Software
- Phases of simulation project

Pause: 15 min.

Part II: Start modeling (theory and praxis)

- Parameters & KPIs
- Processes and BPMN

Introduction of participants

Dr. Tatiana Starostina

1996: Dipl.-Ing. Automated System technology at TU

1999: Dr.-Ing. Mathematical Modeling and Computer Science

Research focus: fuzzy optimization / logistics / transport systems

1999 – 2007: Scientific work at the research department and teaching

2004 – 2005: Guest lecturer TU Bergakademie Freiberg

Since 2007: Scientific staff and Lecturer at HSW / ZHAW in Wädenswil, ICLS

Research areas:

- Simulation & Optimization (Production, warehouse, supply chain management, building planning, traffic and passenger flows)
- Teaching: Computer science, supply chain management, systems engineering, experimental design and evaluation, process engineering and data
- Programming languages: VBA, C++, C#, Python, R, simulation languages (ProModel, Simio, Plant Simulation, Arena, Salabim), other languages for data base and web applications

Contact: stat@zhaw.ch

Marco Brunschwiler

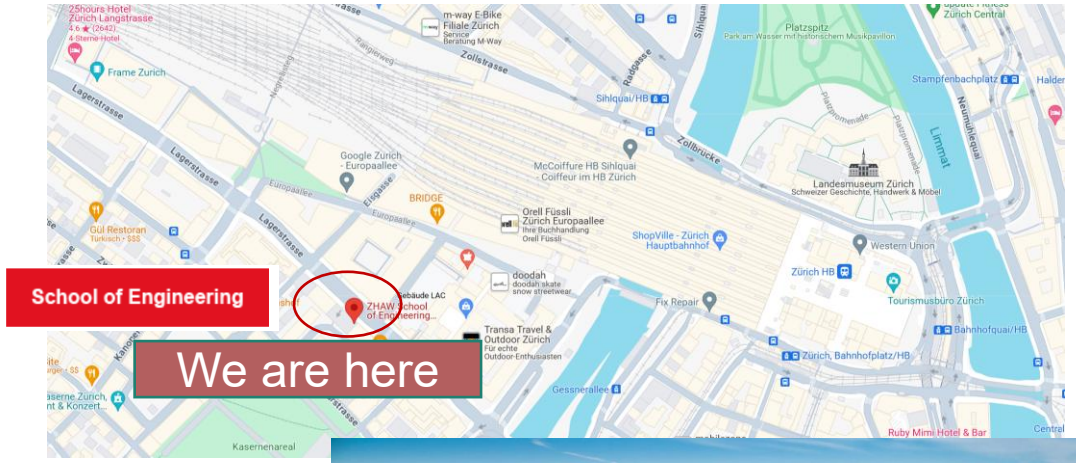
- Business and Economics School (2001 – 2005)
- Various jobs in commercial practice and as a chef (2005 – 2016)
- BSc Food Technology
  ZHAW Wädenswil (2016 – 2020)
- Research Assistant,
  Simulation & Optimization, ZHAW (2019 – 2022)
- MSc Life Sciences, Advanced
  Applied Computational Life Sciences (2020 – 2022)
- Research Associate,
  Simulation & Optimization, ZHAW (from 2022)

- Contact: marco.brunschwiler@zhaw.ch

ZHAW in Wädenswil and our location in Zürich

Departments:

- School of Applied Psychology — Daniela Eberhard
- School of Engineering — Dirk Wilhelm
- School of Health Professions — Andreas Gerber-Grote
- School of Social Work — Frank Wittmann
- School of Applied Linguistics — Daniel Perrin
- School of Life Sciences and Facility Management — Urs Hilber
- School of Architecture, Design and Civil Engineering — Oya Atalay Franck
- School of Management and Law — Reto Steiner



School of Life Sciences and Facility Management

Institutes and centres

- ICBT Institute of Chemistry and Biotechnology
- ICLS Institute of Computational Life Sciences
- IFM Institute of Facility Management
- ILGI Institute of Food and Beverage Innovation
- IUNR Institute of Natural Resource Sciences
- ATV Department Transversalis

Institute of Computational Life Sciences (ICLS)

About 60 employees at ICLS

Our four research centres:

- Bioinformatics
  → develop practical solutions at the interface of biology, medicine and computer sciences
- Cognitive Computing in Life Sciences
  → offer new solutions for the life sciences that are based on the fundamental understanding of human and machine as a learning integrated system
- Computational Health
  → addresses fundamental questions in biology and medicine using computer
- Digital Labs & Production
  → we bring people, spaces and processes together. From mixed reality digita applications and machine-to-machine interfaces, we connect physical with dig analytics
  - **Simulation & Optimization**
  - Data Management & Visualization
  - Edge Computing & Interfaces

https://www.zhaw.ch/en/lsfm/institutes-centres/icls

Research group Simulation & Optimization

Projects:

- Simulation-based plant planning and control
- Increasing efficiency in operational planning
- Production planing und optimization
- Otpimization of logistics and material flow systems
- Organization and construction of order picking systems
- Otpimization of warehouse logistics
- Realization & Optimization of Supply Chain
- Transportation logistics planing
- Traffic and passenger flow simulation
- Strategic and operational simulation and optimization

Project experiences

Simulation & Optimization

- Since 1991 Logistics simulation at HSW / ZHAW
- More than 200 projects

Simulation tools

- ProModel
- Simio
- Salabim
- Plant Simulation (Siemens)
- Arena (Rockwell Automation)

Application areas:

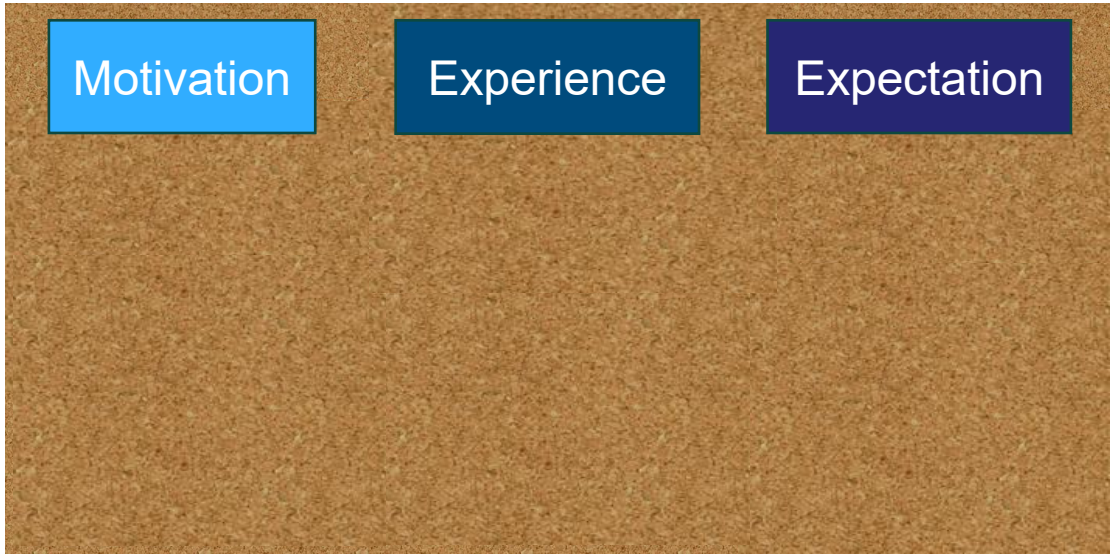
- Industry (chemicals, pharmaceuticals, food, etc.)
- Building planning & facility management (administrative buildings, exhibition centers, theaters, hospitals, vaccination centers, etc.)
- Traffic, transportation
- Other

A part of our project partners

arma suisse
AkzoNobel Tomorrow's Answers Today
ALSTOM
ASTRA
atizo
BAER
ROMER'S HAUSBÄCKEREI AG
Bell
Bischofszell
BÜHLER
CAMILLE BLOCH MON CHOCOLAT SUISSE
chemgineering
coop
Cilag
Emmi
FREY
ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich
Gategourmet a gategroup member
GILGEN LOGISTICS
DIE POST
haco swiss
helbling
hilcona
HUBER+SUHNER
Charles Vögele Switzerland
SPONSER SPORT FOOD
Hero
FOOD ENGINEERING
JOWA
Knorr
e.luterbachag Anlagenbau für die Lackierindustrie
Lindt MAÎTRE CHOCOLATIER SUISSE DEPUIS 1845
Kägi
MAN
METTLER TOLEDO
MICARNA Ein gutes Stück besser - Simplement meilleur
MIDOR Echt Schweiz – Echt gut
mifa ihre marke im mittelpunkt.
MIGROS
V ZUG
MÖVENPICK
Nestlé
NOVARTIS
pistor
P | I | T operations & sourcing entrepreneurs
VALSER
SENS
SBB CFF FFS
REUSS Engineering
Roche
SWISS BIOTECH
WANDER
SIEMENS
SSI SCHÄFER
Stöcklin
stratec biomedical
USM Möbelbausysteme
swissair
valora

Introduction of participants

- Who are you? Short introduction
- Your **motivation**: why you decide to choose this module?
- What **experiences** do you bring to the topic simulation?
- What are your **expectations** of the module?



Learning objectives & Module content

Course «Simulation for Beginners»

2 ECTS

At ZHAW: 3h per week (1/3)

At home: 6h per week (2/3)

You will learn the fundamental concepts of simulation engineering, including

- how to develop and implement simulation models,
- analyze simulation data and
- use simulation software tools.

You will be able after to:

- assess the applicability and usefulness of the simulation method given a specific modelling situation
- list and define typical quantitative objectives for process simulations
- describe and evaluate processes systematically and quantitatively
- implement simple models in a professional simulation environment

Learning objectives

Part „Basics of simulation theory“:

- You know what a process simulation is
- You know if it makes sense to do a simulation study or not
- You know the advantages, disadvantages and benefits of simulation

Part „Simulation project“:

- You know the procedure for implementation a simulation project.
- You can identify and analyze process-related issues in your company or environment.
- You can formulate objectives for problem solving.
- You can describe and define the boundaries of the system to be analyzed based on the problem and objectives.
- You are able to develop simple simulation models on your own.
- You can do experiments, visualize and interpretate results.

Part „Simulation software and simulation model“:

- You understand the basic structure of a simulation model.
- You have gained an initial insight into a simulator and are familiar with the most essential model elements (source, sink, buffer, processing station, moving element).
- You can build small model

Plan of module

Simulation for Beginners			Marco Brunschwiler / Tatiana Starostina	
CW	Date	Topics	Work at home	Time expenditure at home
11	Thursday, 12 March 2026	**Course overview** **Introduction to simulation (Input)** **Introduction to project assignments** **Project assignment 1:** Development of a process flow	**Moodle - Module 1:** What is simulation **Moodle - Module 2:** Installations and first steps **Project assignment 1**	4
12	Thursday, 19 March 2026	**Project example – Vaccination center (20 min)** **Project assignment 2:** conducting experiments and evaluation **Model building:** Build a model from the beginning	**Project Assignments:** Work on Assignments **Moodle - Module 3:** Decide whether to simulate or not	4
13	Thursday, 26 March 2026	**Verification & validation (Input)** **Experiment planning based on research questions (Input)** **Discuss Assignment 1&2 / clarify questions** Model building: Moving resources, state statistics, tally statistics	**Project Assignments:** Work on Assignments **Moodle - Module 4:** Introduction to the simulation study	4
14	Thursday, 2 April 2026	**Easter holidays**	Free	
15	Thursday, 9 April 2026	**Conduct experiments and analyze results** **Discussing results** **Work on Project-Assignments**	**Project Assignments:** Work on Assignments	4
16	Thursday, 16 April 2026	Presentation of results Final review / feedback session	**Project Assignments:** Complete project work	4

Goal: 20 hours

Requirements and Assesment

- **5 times in Zurich** → Compulsory attendance 80% and at the last evening 100%

- **Deadlines:**
  - Modules in Moodle (see Plan)
  - Project 2 documentation (15.04.2026)
  - Presentation of Project 1 (16.04.2026)
- **Assessment:**
  - Presentation → 50%
  - Project documentation (report) → 50%

How we will work

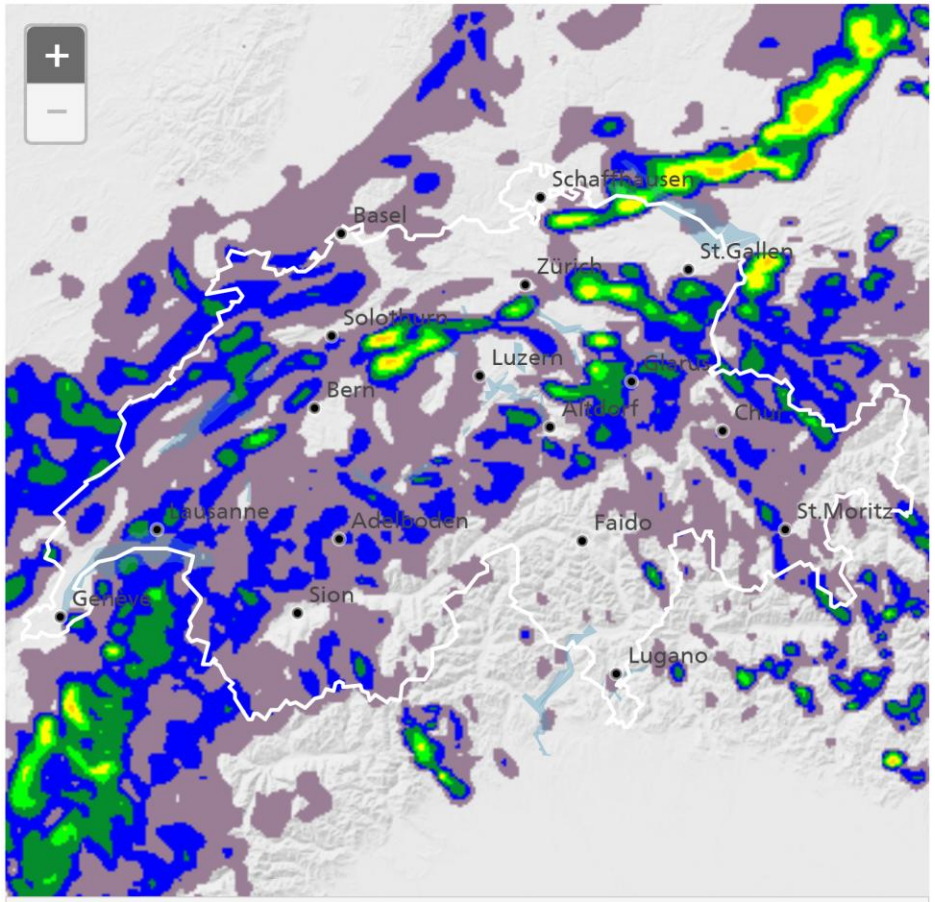
- **Theoretical inputs** and model examples
- **Practical exercises** (group work or independent) in present and at home
  - Theory
  - Process modelling
  - Software Installation → next week
  - Model building
  - Experimentation and interpretation of results
  - Presentation and dokumentation

Questions about the learning objectives, module overview & assessment?

What is Simulation about?

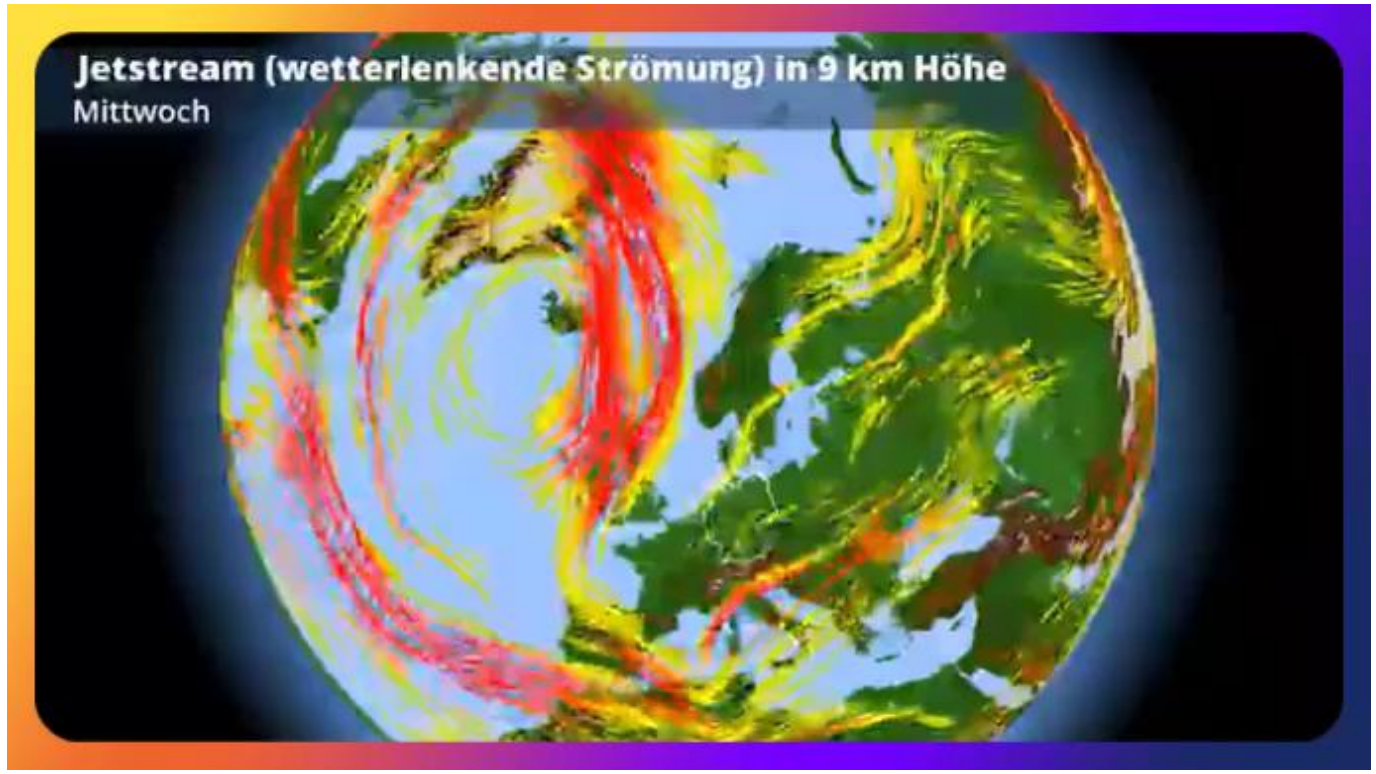
Weather simulation

Weather prediction



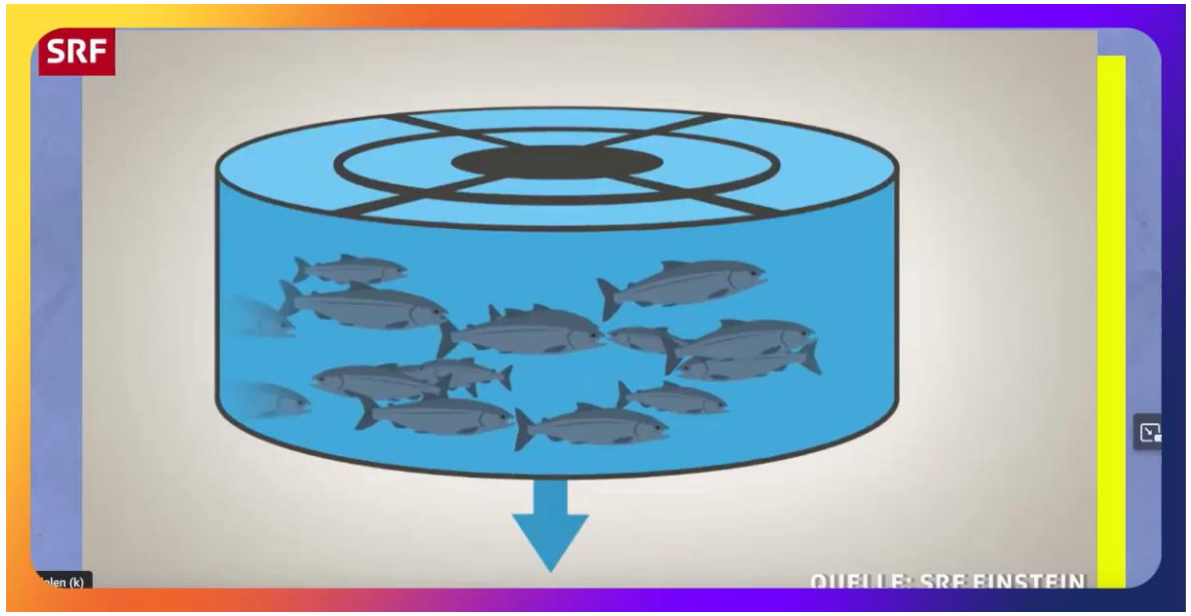
Quelle: http://www.meteoschweiz.admin.ch/

Weather influencing flow



Quelle: https://www.wetter.com/videos/wetter-in-europa/jetstream-5-tages-vorhersage/591b2911cebfc013198b458e

Agent-based simulation of Aquaponic system



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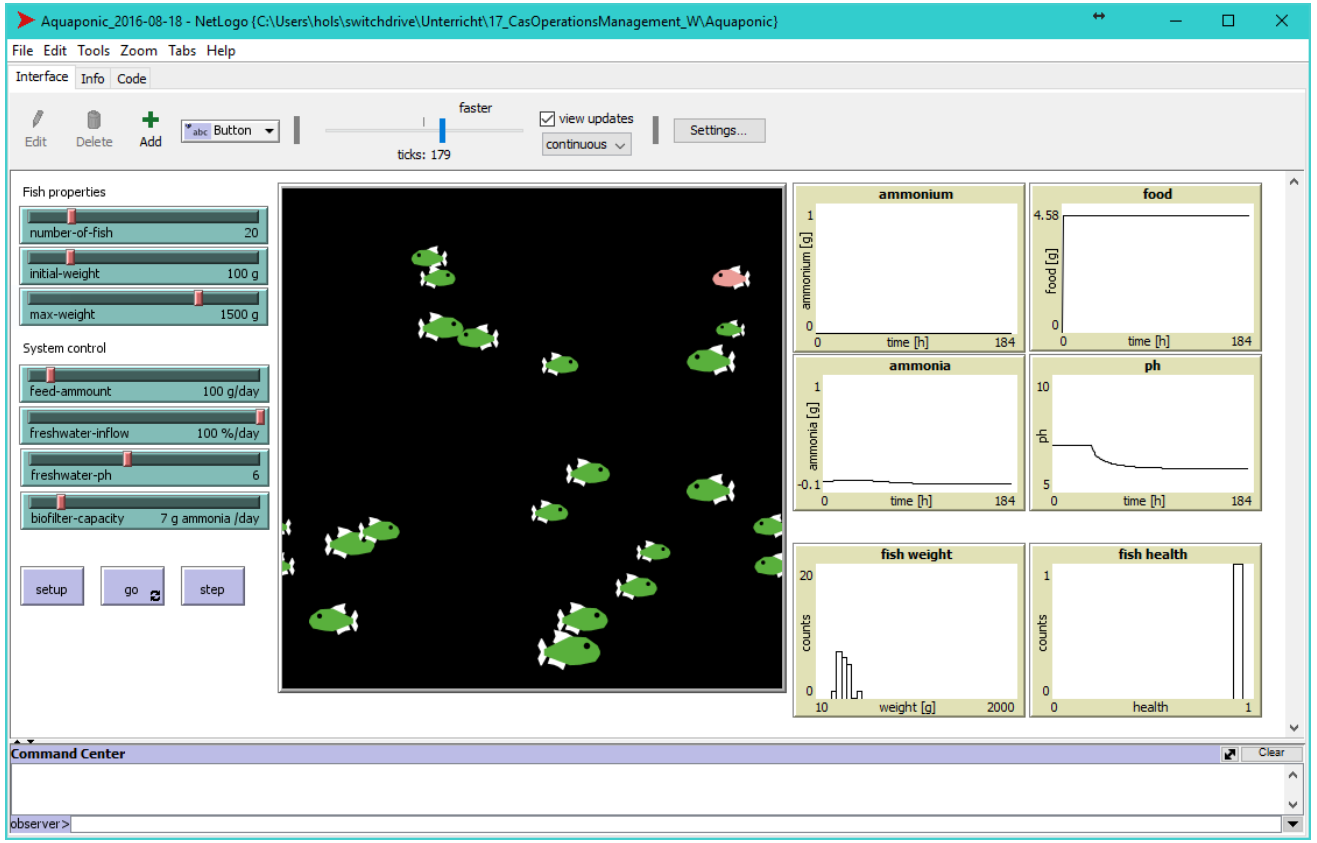
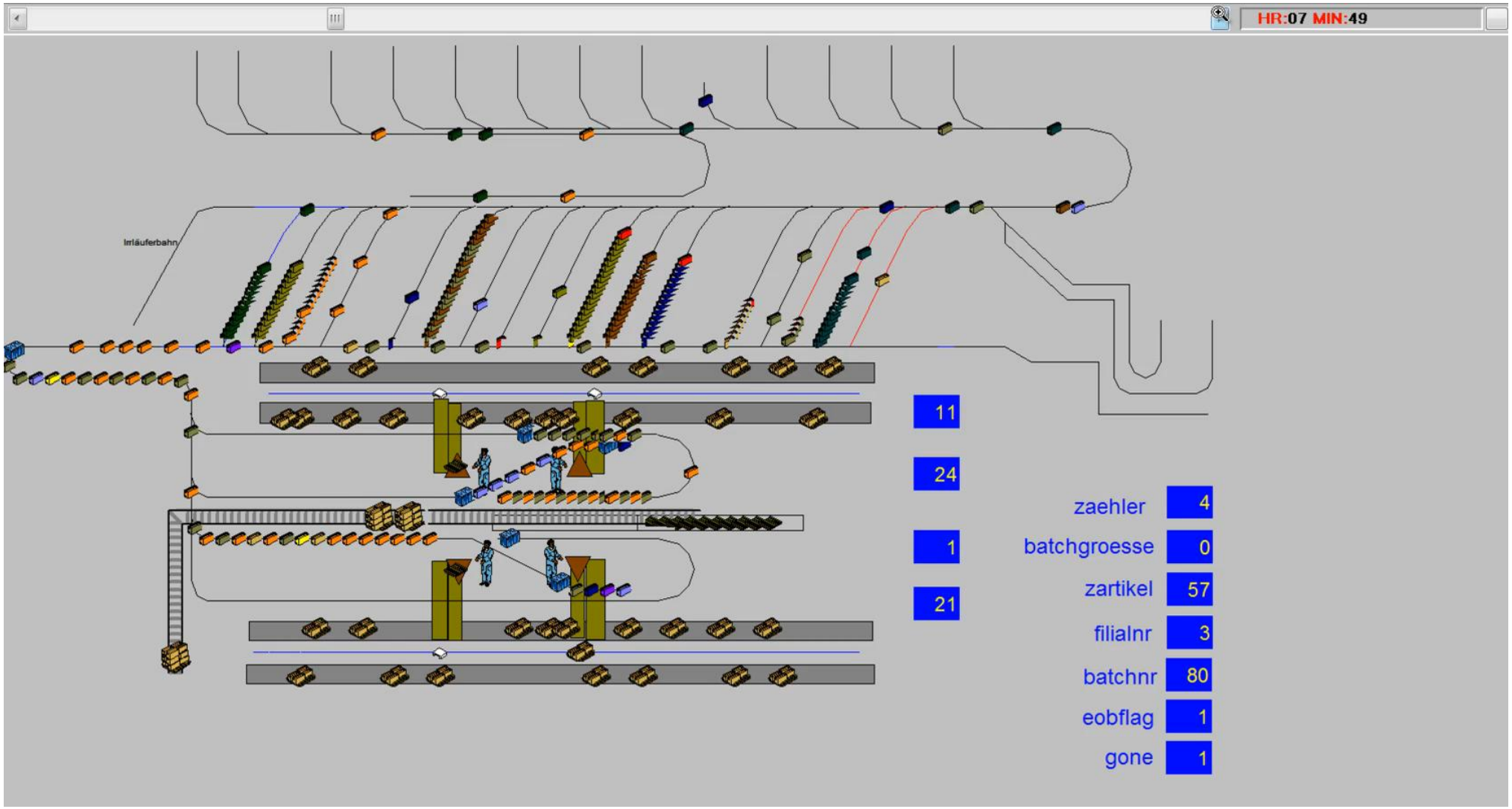


Quelle: IUNR Aquaponics Online-Course and ICLS Simulation & Optimization

Process and Logistics Simulation



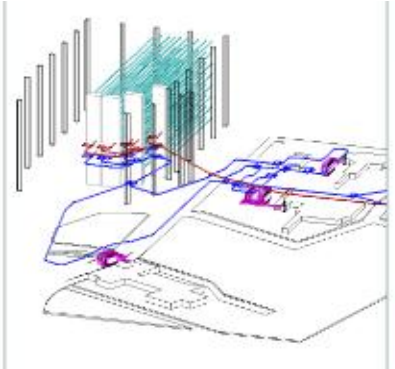
Quelle: ICLS Simulation & Optimization and Migros Genossenschaft Luzern

What is simulation?

- **What do these different simulations have in common?**
  - Reality is "recreated" on the computer.
  - A clearly defined system is described.
  - The underlying model is a model of this system with an explicit level of detail
  - The model must be fed with data.

- **What are the goals of these simulations?**
  - Improve understanding of the system
  - Provide a basis for discussion
  - Proof of performance
  - Predictions
  - Comparison of scenarios and strategies
  - Optimization of processes

Simulate and optimize processes

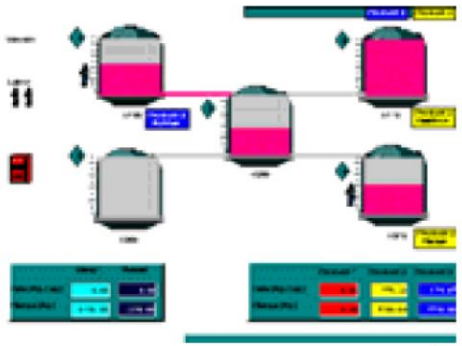


Simulation

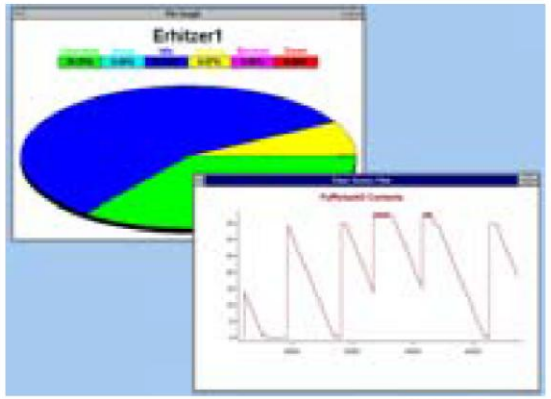
Real system

Modeling
Abstraction

Simulation model



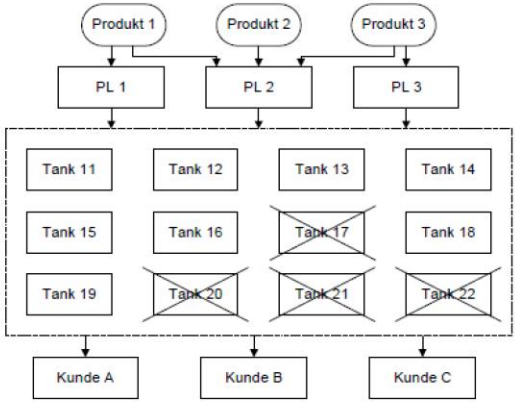
Experiments



Results / Insights

Knowledge transfer
Interpretation

Conclusions



Implementation

Our focus:
Process simulation

Simulation combines the possibilities of tables calculations (e.g. in Excel) with the planning precision of CAD programs in 3D space, **taking time into account**.

Quelle: M. Burkhard (2019) SCM Supply Chain Management. Grundlagen. ZHAW Wädenswil

Why is simulation done? One of several reasons

- Systems are complex and cannot be observed directly or testing them is expensive.
  → Simulations make it possible to make processes more efficient, sustainable, and economical before they are implemented in practice.

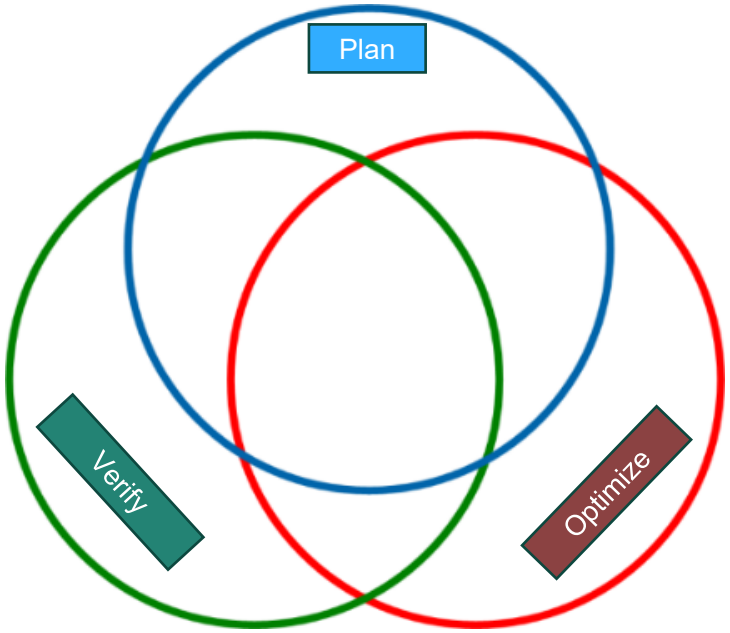
Reasons for simulation studies and benefits

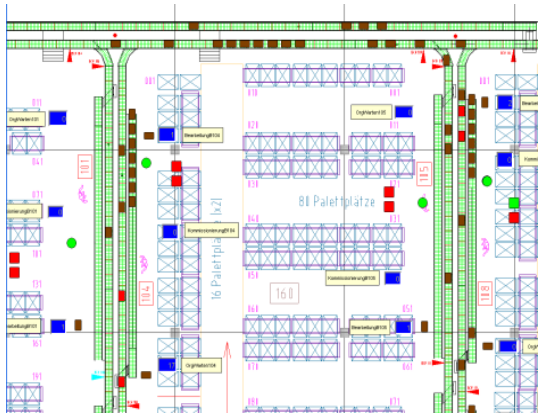
Reasons:

- Proof of performance
- Investment decision
- Process optimization (strategic)
- Production optimization or planning (operational)
- Simulation as a discussion or sales tool

Results:

- Insights about performance limits
- Individual experiments and statistical evaluations
- Visualization





Benefits:

- Increased safety
  - Proof of performance minimize planning risk
- System understanding
  - Understanding dynamic dependencies
  - Visualization
- Increased flexibility & cost reduction
  - Optimized plant configuration
  - Optimized plant control
  - Optimized resource allocation

Key aspects and limitations

Key aspects are:

- Definition of the system and its boundaries
- Level of details (degree of abstraction)
- Modeling of processes within the system
- Modeling of dependencies on external factors
- Data basis

Limitations:

- **System & system boundaries**
  The modeled system does not contain the entire universe.
- **Level of detail**
  The model has a certain level of detail and cannot resolve more than what it was created for.
- **Data**
  The quality of the results depends essentially on the quality of the data.

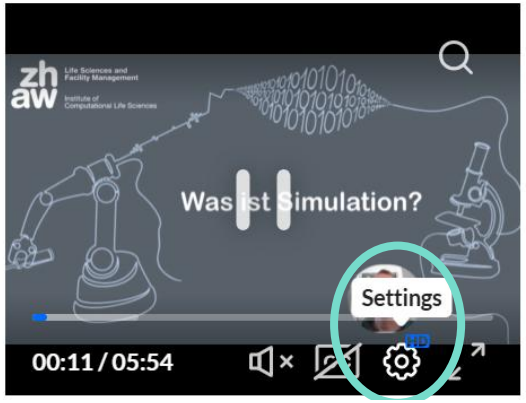
Questions about Simulation?

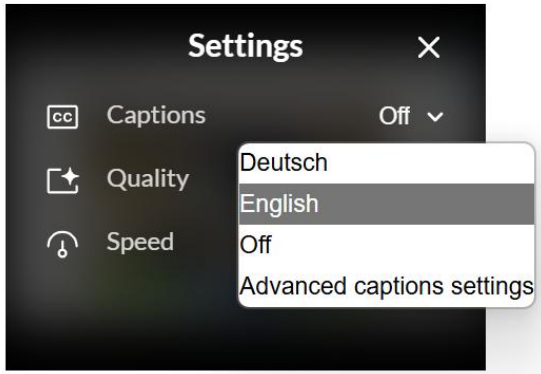
Additional information and learning control

- Additional information and learning control in Moodle course
- **Module 1: at home till next Thursday**

Videos are in German, but have English subtitel

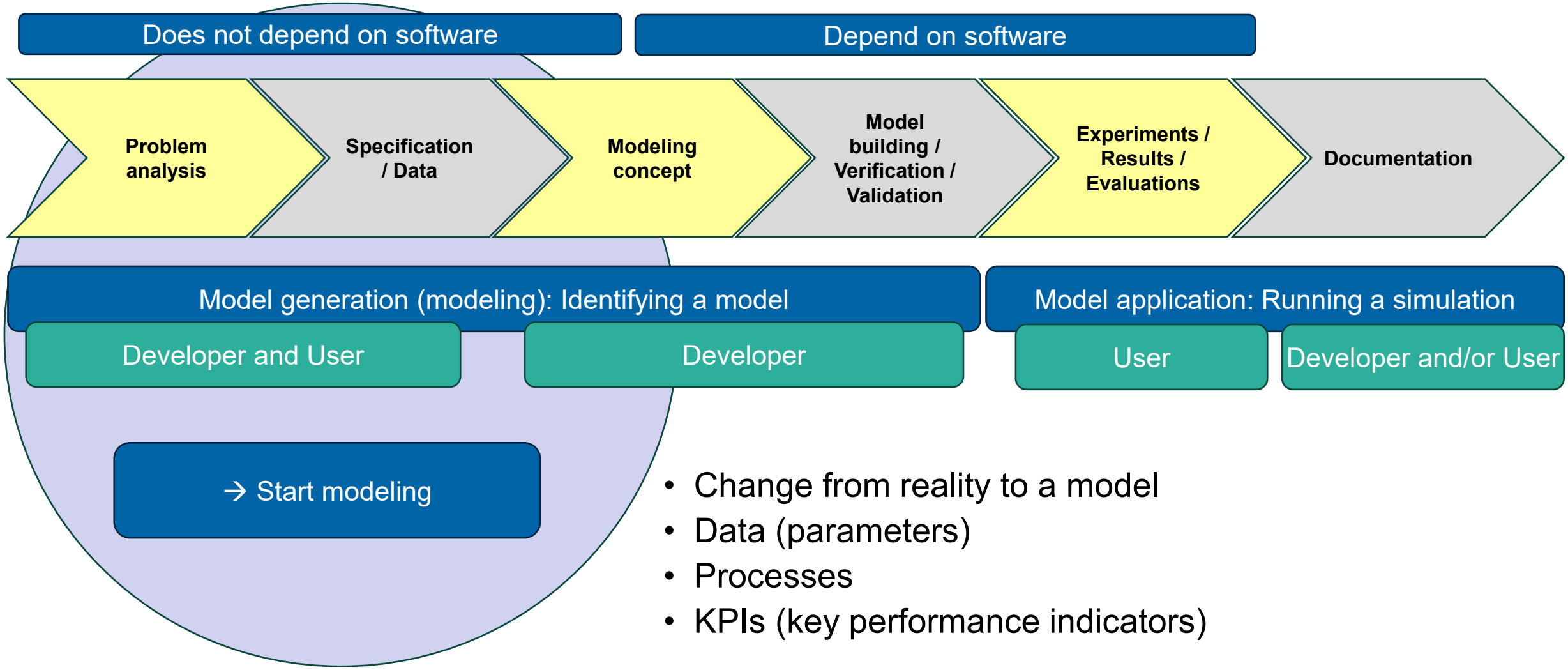
To switch on:





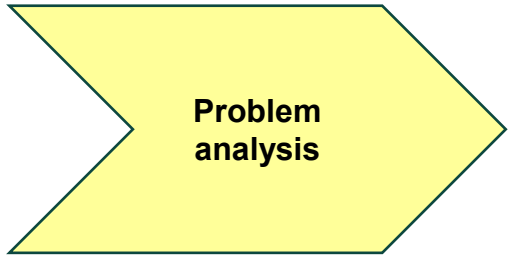
Short information about simulation project structure and first steps

Phases of simulation project



- Change from reality to a model
- Data (parameters)
- Processes
- KPIs (key performance indicators)

First steps in the project



1. What is a problem about?

   What is important and what not?

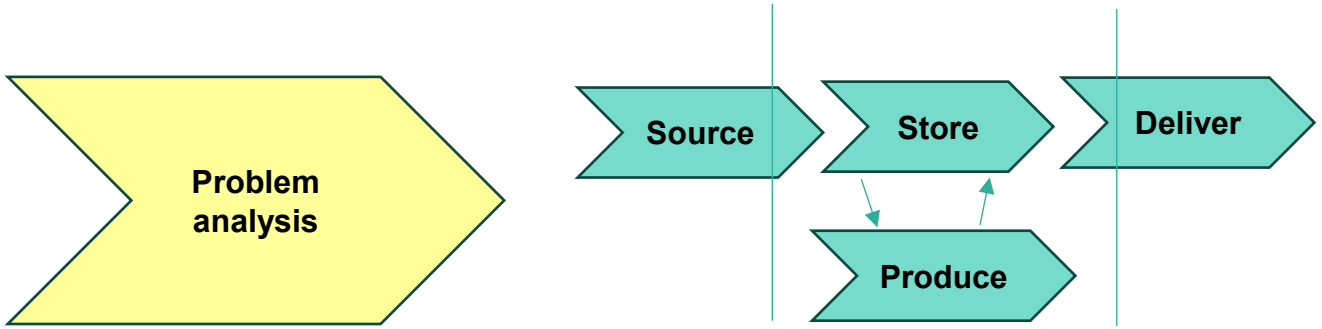
   What are objectives of simulation?

2. Define a system and the boundaries of the system

3. What abstractions is enough for us?

   Set a level of detail (how deep is our investigation?)

First steps in the project and one example of Warehouse modeling



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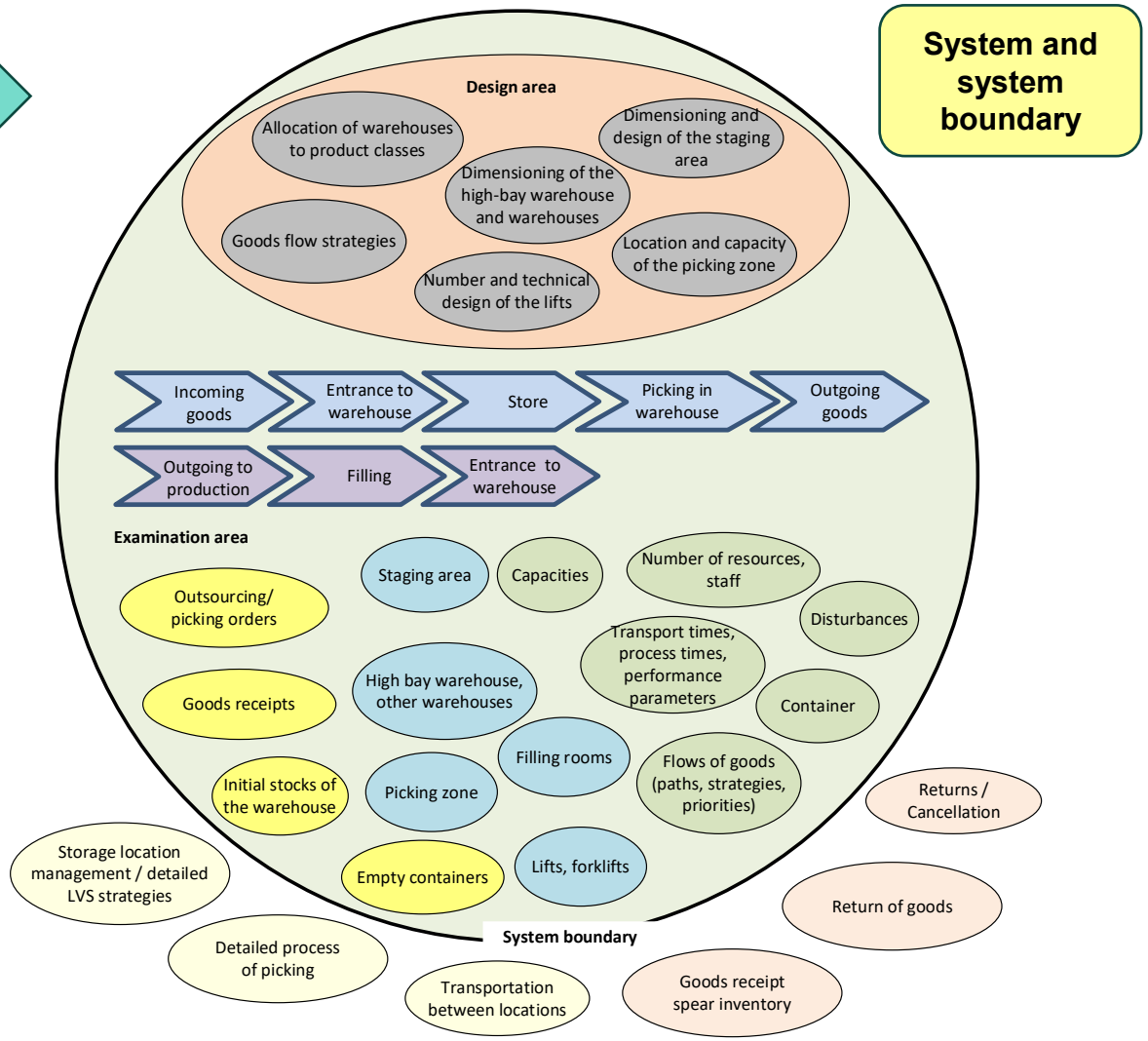
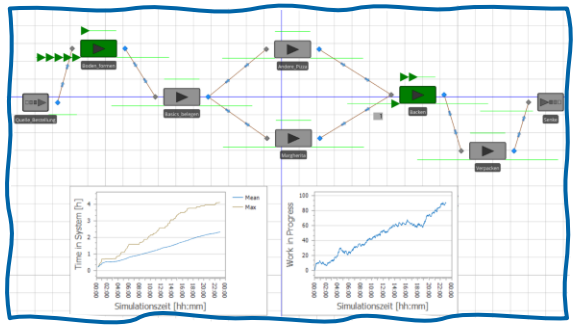


From real situation to the model

Does not depend on software

Problem analysis → **Specification / Data** → **Modeling concept** → **Model building / Verification / Validation**



First modeling steps → without simulation software

Depend on software

Simulation software and model → next week